A SEPARATED FINITE TYPE ALGEBRAIC SPACE OVER AN AFFINE BASE ADMITS A PROPER SURJECTIVE COVER BY AN H-QUASI-PROJECTIVE SCHEME (THE WEAK CHOW LEMMA)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Boundedness of the geometric fibres of a surjective etale cover, and the stratification where the cover becomes finite of the minimal degree. Decent Spaces, labels decent-spaces-lemma-bounded-fibres (invoked at proof line 3510) and decent-spaces-lemma-quasi-compact-reasonable-stratification (invoked at proof line 3514). They produce an integer d bounding the number of geometric fibre points of the surjective etale cover $W \to X$, and, for d minimal, a nonempty open $U \subset X$ over which the cover is finite etale of degree d.
- The diagonal of a separated etale morphism is an open and closed immersion. Morphisms of Spaces, labels spaces-morphisms-lemma-etale-unramified and spaces-morphisms-lemma-diagonal-unramified-morphism (invoked at proof lines 3524 and 3525). It makes the diagonals open and closed subschemes of the d-fold fibre product, hence the complement V of all diagonals is a quasi-compact scheme.
- Every affine scheme of finite type embeds as an open subscheme of an H-projective scheme. Morphisms, labels morphisms-lemma-quasi-affine-finite-type-over-S and morphisms-lemma-H-quasi-projective-open-H-projective (invoked at proof lines 3533 and 3534). It provides the open immersion $W \subset Y$ with Y H-projective over A into which the construction is placed.
- The refined valuative criterion for universal closedness, and lifting points through the flat cover after extending the valuation ring. Morphisms of

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Spaces, labels spaces-morphisms-lemma-refined-valuative-criterion-universally-closed (invoked at proof line 3576) and spaces-morphisms-lemma-lift-valuation-ring-through-flat-morphism (invoked at proof line 3596). The criterion reduces properness of $X' \to X$ to lifting valuation ring diagrams, and the lifting lemma supplies $w : \Spec(R) \to W$ over $\Spec(R) \to X$ after extending R.

- The valuative criterion of properness for the projective morphisms g_i. Morphisms, label morphisms-lemma-locally-projective-proper and Schemes, label schemes-proposition-characterize-universally-closed (invoked at proof lines 3606-3608). They lift w to a morphism $z : \Spec(R) \to Z$ whose image lands in $g_i^{-1}(W) \subset X'$, completing the verification of the claim.

Source reference. Original lemma 03JX: sources/stacks-chow/tags.txt, source lines 4655–4655. Navigation pointer for \ref{decent-spaces-lemma-bounded-fibres} used inside the counted span (line 3510); the argument that consumes it is retained in the span.

Source reference. Original lemma 07S8: sources/stacks-chow/tags.txt, source lines 9893–9893. Navigation pointer for \ref{decent-spaces-lemma-quasi-compact-reasonable-stratification} used inside the counted span (line 3514); the argument that consumes it is retained in the span.

Source reference. Original lemma 01WA: sources/stacks-chow/tags.txt, source lines 2475–2475. Navigation pointer for \ref{morphisms-lemma-H-quasi-projective-open-H-projective} used inside the counted span (line 3534); the argument that consumes it is retained in the span.

Source reference. Original lemma 01WC: sources/stacks-chow/tags.txt, source lines 2477–2477. Navigation pointer for \ref{morphisms-lemma-locally-projective-proper} used inside the counted span (line 3609); the argument that consumes it is retained in the span.

Source reference. Original lemma 04II: sources/stacks-chow/tags.txt, source lines 5904–5904. Navigation pointer for \ref{morphisms-lemma-quasi-affine-finite-type-over-S} used inside the counted span (line 3533); the argument that consumes it is retained in the span.

Source reference. Original lemma 01RG: sources/stacks-chow/tags.txt, source lines 2301–2301. Navigation pointer for \ref{morphisms-lemma-quasi-compact-immersion} used inside the counted span (line 3545); the argument that consumes it is retained in the span.

Source reference. Original section 01W7: sources/stacks-chow/tags.txt, source lines 2472–2472. Navigation pointer for \ref{morphisms-section-projective} used inside the counted span (line 3548); the argument that consumes it is retained in the span.

Source reference. Original proposition 01KF: sources/stacks-chow/tags.txt, source lines 2048–2048. Navigation pointer for \ref{schemes-proposition-characterize-universally-closed} used inside the counted span (line 3611); the argument that consumes it is retained in the span.

Source reference. Original lemma 05W1: sources/stacks-chow/tags.txt, source lines 7570–7570. Navigation pointer for \ref{spaces-morphisms-lemma-diagonal-unramified-morphism} used inside the counted span (line 3525); the argument that consumes it is retained in the span.

Source reference. Original lemma 084N: sources/stacks-chow/tags.txt, source lines 10328–10328. Navigation pointer for \ref{spaces-morphisms-lemma-equality-of-morphisms}

used inside the counted span (line 3557); the argument that consumes it is retained in the span.

Source reference. Original lemma 06CR: sources/stacks-chow/tags.txt, source lines 8155–8155. Navigation pointer for \ref{spaces-morphisms-lemma-etale-unramified} used inside the counted span (line 3524); the argument that consumes it is retained in the span.

Source reference. Original lemma 089F: sources/stacks-chow/tags.txt, source lines 10495–10495. Navigation pointer for \ref{spaces-morphisms-lemma-lift-valuation-ring-through-flat-morphism} used inside the counted span (line 3597); the argument that consumes it is retained in the span.

Source reference. Original lemma 089G: sources/stacks-chow/tags.txt, source lines 10496–10496. Navigation pointer for \ref{spaces-morphisms-lemma-refined-valuative-criterion-universally-closed} used inside the counted span (line 3577); the argument that consumes it is retained in the span.

Lemma 0.1. *Let A be a ring. Let X be an algebraic space over* $\mathrm{Spec}(A)$ *whose structure morphism $X \to \mathrm{Spec}(A)$ is separated of finite type. Then there exists a proper surjective morphism $X' \to X$ where X' is a scheme which is H-quasi-projective over* $\mathrm{Spec}(A)$.

Proof. Let W be an affine scheme and let $f : W \to X$ be a surjective étale morphism. There exists an integer d such that all geometric fibres of f have $\leq d$ points (because X is a separated algebraic hence reasonable, see Decent Spaces, Lemma 03JX). Picking d minimal we get a nonempty open $U \subset X$ such that $f^{-1}(U) \to U$ is finite étale of degree d, see Decent Spaces, Lemma 07S8. Let

$$V \subset W \times_X W \times_X \ldots \times_X W$$

(d factors in the fibre product) be the complement of all the diagonals. Because $W \to X$ is separated the diagonal $W \to W \times_X W$ is a closed immersion. Since $W \to X$ is étale the diagonal $W \to W \times_X W$ is an open immersion, see Morphisms of Spaces, Lemmas 06CR and 05W1. Hence the diagonals are open and closed subschemes of the quasi-compact scheme $W \times_X \ldots \times_X W$. In particular we conclude V is a quasi-compact scheme. Choose an open immersion $W \subset Y$ with Y H-projective over A (this is possible as W is affine and of finite type over A; for example we can use Morphisms, Lemmas 04II and 01WA). Let

$$Z \subset Y \times_A Y \times_A \ldots \times_A Y$$

be the scheme theoretic image of the composition $V \to W \times_X \ldots \times_X W \to Y \times_A \ldots \times_A Y$. Observe that this morphism is quasi-compact since V is quasi-compact and $Y \times_A \ldots \times_A Y$ is separated. Note that $V \to Z$ is an open immersion as $V \to Y \times_A \ldots \times_A Y$ is an immersion, see Morphisms, Lemma 01RG. The projection morphisms give d morphisms $g_i : Z \to Y$. These morphisms g_i are projective as Y is projective over A, see material in Morphisms, Section 01W7. We set

$$X' = \bigcup g_i^{-1}(W) \subset Z$$

There is a morphism $X' \to X$ whose restriction to $g_i^{-1}(W)$ is the composition $g_i^{-1}(W) \to W \to X$. Namely, these morphisms agree over V hence agree over $g_i^{-1}(W) \cap g_j^{-1}(W)$ by Morphisms of Spaces, Lemma 084N. Claim: the morphism $X' \to X$ is proper.

If the claim holds, then the lemma follows by induction on d. Namely, by construction X' is H-quasi-projective over $\mathrm{Spec}(A)$. The image of $X' \to X$ contains the

open U as V surjects onto U. Denote T the reduced induced algebraic space structure on $X \setminus U$. Then $T \times_X W$ is a closed subscheme of W, hence affine. Moreover, the morphism $T \times_X W \to T$ is étale and every geometric fibre has $< d$ points. By induction hypothesis there exists a proper surjective morphism $T' \to T$ where T' is a scheme H-quasi-projective over $\mathrm{Spec}(A)$. Since T is a closed subspace of X we see that $T' \to X$ is a proper morphism. Thus the lemma follows by taking the proper surjective morphism $X' \amalg T' \to X$.

Proof of the claim. By construction the morphism $X' \to X$ is separated and of finite type. We will check conditions (1) – (4) of Morphisms of Spaces, Lemma 089G for the morphisms $V \to X'$ and $X' \to X$. Conditions (1) and (2) we have seen above. Condition (3) holds as $X' \to X$ is separated (as a morphism whose source is a separated algebraic space). Thus it suffices to check liftability to X' for diagrams

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & V \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \longrightarrow & X \end{array}$$

where R is a valuation ring with fraction field K. Note that the top horizontal map is given by d pairwise distinct K-valued points $w_1, \ldots, w_d$ of W. In fact, this is a complete set of inverse images of the point $x \in X(K)$ coming from the diagram. Since $W \to X$ is surjective, we can, after possibly replacing R by an extension of valuation rings, lift the morphism $\mathrm{Spec}(R) \to X$ to a morphism $w : \mathrm{Spec}(R) \to W$, see Morphisms of Spaces, Lemma 089F. Since $w_1, \ldots, w_d$ is a complete collection of inverse images of x we see that $w|_{\mathrm{Spec}(K)}$ is equal to one of them, say w_i. Thus we see that we get a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & Z \\ \downarrow & & \downarrow{\scriptstyle g_i} \\ \mathrm{Spec}(R) & \xrightarrow{w} & Y \end{array}$$

By the valuative criterion of properness for the projective morphism g_i we can lift w to $z : \mathrm{Spec}(R) \to Z$, see Morphisms, Lemma 01WC and Schemes, Proposition 01KF. The image of z is in $g_i^{-1}(W) \subset X'$ and the proof is complete. □